Silverline House — 25-Storey Residential HRB

# Smoke Extraction Strategy Note

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# 1 Introduction

This Fire Strategy Report has been prepared to accompany the Gateway 2 application for Building Control Approval for Silverline House, a 25-storey residential higher- risk building. The strategy has been developed in accordance with Approved Document B (Fire Safety) Volumes 1 and 2, BS 9999:2017, and BS 9991:2015. The report identifies and justifies all elements of the fire safety design including means of escape, passive fire protection, active fire suppression, fire detection, and fire service access. The strategy is supported by evacuation simulations carried out in accordance with PD 7974-6 and confirmatory computational fluid dynamics (CFD) modelling of the smoke extraction system. This note describes the smoke extraction strategy for the common areas, protected corridors, and car parking levels.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

## 2 Smoke Control System Design

Smoke extraction for the protected corridors is achieved by natural ventilation through AOV (automatic opening vent) panels at the head of each staircase, supplemented by mechanical smoke extraction from the lift lobbies using dedicated axial flow fans sized at 10 air changes per hour. The system is activated on detection of fire by the fire alarm system. An automatic fire detection and alarm system has been designed in accordance with BS 5839-1 Category L2 for the common areas and Category LD2 for the individual apartments. Optical smoke detectors are installed in all protected corridors, lobbies, and staircase enclosures. Combination heat/smoke detectors are installed in apartment entrance halls. The system is connected to a central addressable control panel located in the ground-floor security room with a sub-panel in the Level 13 plant room. Manual call points are installed at each staircase landing on every floor and at each exit door at ground level. The detection system is interfaced with the building management system (BMS) to allow integration with smoke control, lift recall, and door hold-open release functions.

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## 3 CFD Analysis Summary

Computational fluid dynamics (CFD) modelling has been carried out using FDS (Fire Dynamics Simulator) v6.7 to validate the smoke extraction strategy. The modelling confirms that tenable conditions are maintained in the protected corridors for a minimum of 5 minutes from ignition, providing sufficient time for occupant evacuation. The fire alarm system is configured to provide a single-stage alarm throughout the building on detection of fire. The alarm signal is transmitted to the fire service via an automatic alarm transmission system (ATS) with a dedicated monitored line in accordance with BS 5839-1 clause 25.2. The alarm system is supplied from the building's UPS system providing a minimum 72-hour standby capacity. The main control panel provides a graphical display of the building floor plans indicating the location of activated detectors to assist the fire service on arrival. A cause-and-effect matrix is included in Appendix C of this report, showing the response of all interfaced systems (smoke control, lifts, access control, hold-open devices) on activation of each detector zone.

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